

PERRY CHEUNG

Senior Frontend Engineer (React / TypeScript, Next.js, Node.js)

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Vancouver, BC, Canada

Professional Summary

- **Senior Frontend Engineer** specializing in scalable SaaS systems and complex workflows, with 10+ years of experience building performance-critical applications. Expert in **React**, **TypeScript**, and **Next.js**, with a strong track record delivering data-driven platforms across **SaaS**, **FinTech**, and **enterprise** environments.
- Led frontend modernization at **TechPOS (Canada Retail POS SaaS)** and engineered **GovTech CMS (Simplicity CMS)** infrastructure for BC municipalities at **Radical I/O**. Previously delivered enterprise-grade trading platforms at **Haitong International Securities** and **Serai (HSBC Group)** in Hong Kong.
- Full-stack capability in **Node.js** and **NestJS** to own features end-to-end. Track record of measurable impact: **30%+ performance gains**, **40% onboarding reduction**, **75% faster form development**.
- **Google UX Design certified**, with a strong focus on accessible, user-centred interfaces.

Technical Skills

Frontend Architecture: React, TypeScript, Next.js, Redux/Zustand, AgGrid, D3.js, Storybook, HTML5/CSS3 (SASS)

Backend & APIs: Node.js, NestJS, C# (.NET), API Design (REST/GraphQL), PostgreSQL, MongoDB

AI & Tools: Claude, OpenAI API (LLM-assisted development workflows), Cursor

Cloud & DevOps: Azure, AWS (EC2/S3), Docker, Kubernetes, CI/CD (Jenkins, GitHub Actions), Git

Testing & Quality: Cypress (E2E), Jest, Unit & Integration Testing

UX & Accessibility: WCAG Accessibility Standards, Figma

Working Experience

Senior Frontend Engineer (UX/UI Developer)

Oct 2024 - Jan 2026

TechPOS · Canada's #1 Cannabis Retail SaaS · 1,000+ dispensaries

Vancouver, BC, Canada

- **Legacy Modernization:** Owned the frontend modernisation strategy for an **8-year legacy POS platform** — designed an incremental migration path that delivered zero-downtime cutover with full **Cypress E2E** coverage protecting production.
- **Performance Engineering:** Architected a high-performance reporting engine handling 10,000+ records with zero UI lag, using AgGrid virtualisation and intelligent data summarisation — replacing a system merchants had stopped trusting.
- **Product Impact:** Designed and shipped a **Smart Setup onboarding** workflow with a predictive UI suggestion engine, **cutting merchant onboarding time by 40%** and **reducing support escalations**.
- **Next.js Migration:** Improved organic discoverability and achieved a **30% boost in page load speed** by orchestrating a legacy codebase revamp to **Next.js**, leveraging **SSR/SSG** and strategic code-splitting for **optimal SEO performance**.

Stack: React, Next.js, TypeScript, Redux, Material UI, AgGrid, REST APIs, PostgreSQL, Storybook, Cypress, C#, Azure

Senior Frontend Engineer

Mar 2023 - Feb 2024

Radical I/O Company · Vancouver GovTech · Simplicity CMS for BC municipalities

Vancouver, BC, Canada

- **Workflow Efficiency:** Reduced government permit processing time by 50% by **architecting a dynamic, JSON-driven form engine** with automated validation workflows.
- **Low-Code Empowerment:** Eliminated the need for engineering intervention in content updates by building **high-complexity CMS authoring tools**, enabling non-technical users to **manage dynamic layouts and media** independently
- **Scalable SaaS Architecture:** Engineered multi-tenant SaaS CMS platforms using **React** and **TypeScript**, implementing a modular architecture that maximized code reuse across enterprise projects.

Stack: React, TypeScript, Svelte, Node.js, PostgreSQL, MongoDB, REST APIs, Jest, Jenkins, AWS (EC2, S3), Git

Lead Engineer (Independent Project)

Feb 2024 – Present

Sniff & Frolic · Headless E-commerce platform · POS

Vancouver, BC, Canada

- **Headless Architecture:** Architected a **headless e-commerce platform** using **Next.js**, integrating WooCommerce APIs and transitioning to a custom **NestJS + PostgreSQL** backend to improve performance, data control, and system flexibility
- **Performance Engineering:** Designed product, cart, and checkout systems with **SSR/SSG optimization** for **performance** and **SEO**
- **POS System:** Built a **modular POS system** using **React**, **TypeScript**, and **Zustand** to support real-time sales operations and multi-role workflows

- **AI Systems:** Implemented **Claude API-powered product discovery** and **SEO metadata generation** with constrained context design, integrated into e-commerce workflows

Stack: React, Next.js, TypeScript, Zustand, Node.js (NestJS), REST APIs, PostgreSQL, Storybook, Playwright, Vercel, Supabase

Lead Frontend Developer & UX Architect

Sep 2020 - Feb 2023

Haitong International Securities Group Ltd · One of China's largest investment banks

Hong Kong, China / Vancouver (Remote)

- **Scalable Enterprise Foundations:** Solely designed an enterprise-grade React architecture from the ground up over 6 months — OAuth, RBAC, and granular permissions — adopted across 2 trading platforms and extended by additional teams to 4 more business systems without architectural changes; retained as architectural owner throughout.
- **Form Engine:** Designed and implemented a **configuration-driven workflow system** supporting **validation, conditional logic**, and role-based access across multiple business workflows
- **Rapid Development Framework:** Reduced form development time by **75%** by delivering a process-driven, dynamic framework supporting **200+ configurable trading workflows**.
- **Engineering Leadership:** Led a team of **5 frontend engineers** to deliver full digitization of **200+ workflows** within a 2-year timeline — overseeing code reviews, architectural guidance, and end-to-end quality standards.

Stack: React, Material UI, REST APIs, PostgreSQL, Jest, Docker, Kubernetes, Jenkins, Git, Vue

Frontend Developer / UI Developer

Dec 2019 - Sep 2020

Serai Limited · HSBC's first non-banking tech venture · B2B trade platform

Hong Kong, China

- **Data-Informed Decision Making:** Developed advanced **D3.js data-visualization** modules that surfaced supplier analytics, enabling internal teams to interpret complex supply-chain metrics efficiently. Led 2 frontend engineers across both products.
- **High-Performance UI:** Integrated **REST and GraphQL APIs** to power data-intensive fintech dashboards, ensuring low-latency performance for complex operational workflows.

Stack: React, Redux, Material UI, D3.js, Node.js, REST APIs, GraphQL, PostgreSQL, MongoDB, Jest, Cypress, AWS

Senior Frontend Developer

Dec 2017 - Sep 2019

Fayfay.com Limited · Travel Ecommerce · MoUs with Da Nang, Hue & HCMC Tourism Boards

Hong Kong, China

- **Marketing Efficiency Boost:** Reduced marketing implementation time by **70%** by delivering a library of reusable campaign components and a custom internal CMS for rapid product launches.
- **Full-Cycle E-commerce Scaling:** Built and scaled a comprehensive e-commerce engine, including secure **Stripe payment integrations**, checkout flows, and complex promotion modules.

Stack: React, Redux, Node.js, REST APIs, PostgreSQL, MongoDB, Stripe, Google Analytics, AWS

Web Developer

Nov 2015 - Dec 2017

The Hong Kong Jockey Club · HK's largest taxpayer · millions of users across racing, lottery & sports betting

Hong Kong, China

Developed high-traffic e-commerce and campaign platforms (hkjc.com) using ASP.NET and modern JavaScript.

Senior Web Developer

Sep 2014 - Oct 2015

Isobar Hong Kong Limited · APAC Digital Agency of the Year · Peninsula Hotels · Audi · UNIQLO

Hong Kong, China

Delivered high-profile digital campaigns for global brands including **Audi** and **UNIQLO**; managed junior developers.

Education

Master's in E-commerce,

The Hong Kong Polytechnic University (2019)

Bachelor of Arts in Applied and Media Arts,

The Hong Kong Polytechnic University (2013)

Certificate

Google **UX Design Professional** Certificate

Google **Digital Marketing & E-commerce** Certificate